The working process: inputs to a dollar price

Allowed inputs → engineered features → log-price model → dollar prediction

1 SELECT ALLOWED INPUTS

Original property, location, listing date and remarks

2 ENGINEER FEATURES

Area midpoints · property ratios · fixed text cues

3 FIT CATEGORY ENCODING

Fit only within the training fold
Encoded fields remain categorical

4 LEARN LOG SALE PRICE

Histogram gradient boosting

Group numeric values into bins; learn missing-value routes

Add trees that correct remaining log-price errors

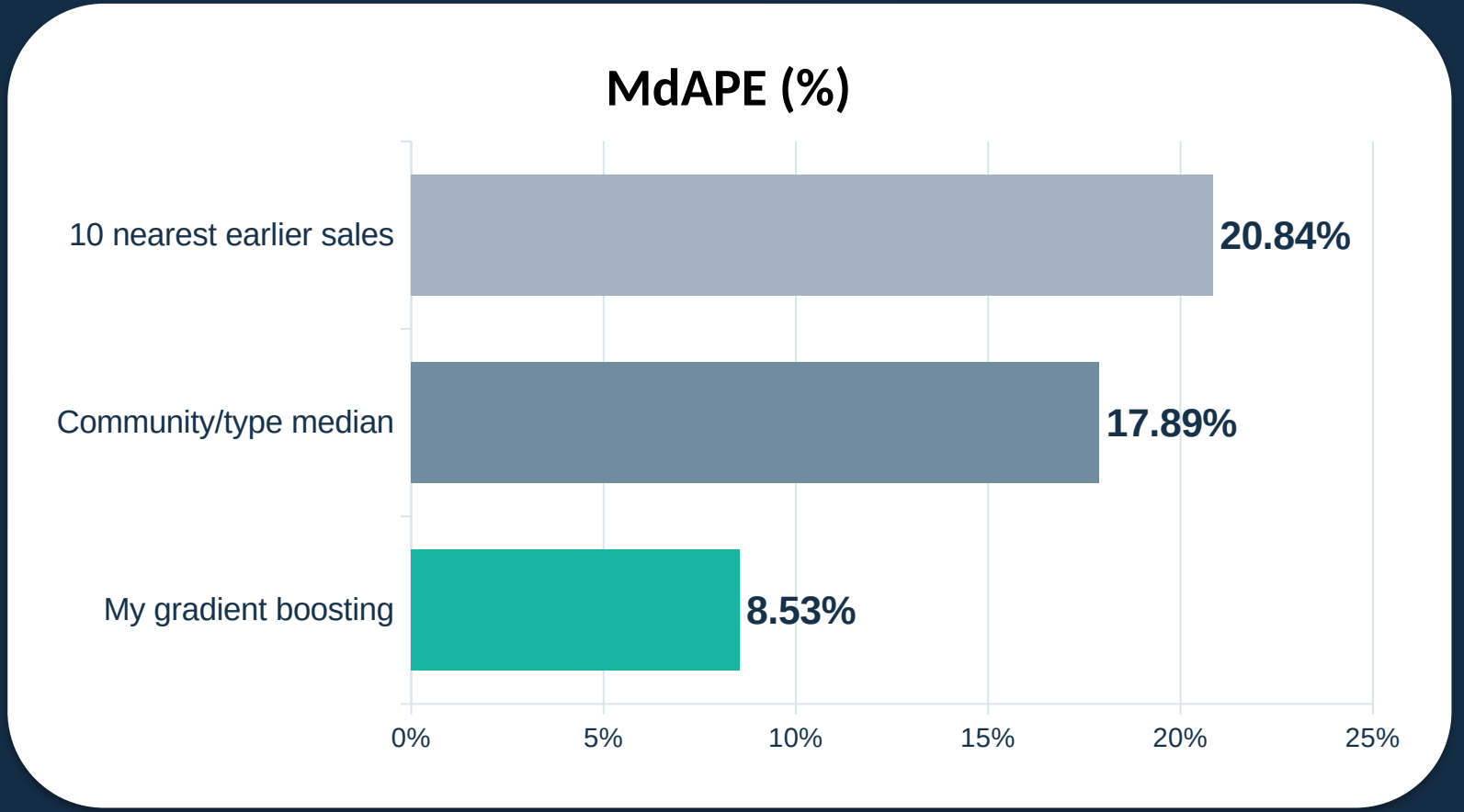
5 COMBINE AND CONVERT TO DOLLARS

price = exp(predicted log price)

Excluded: list price • post-sale fields • AVMs • IDs

Validate on later sales, then compare error

For each month: earlier sales → held-out predictions → errors against original sale prices



8.53%

Median absolute % error

11.00%

Mean absolute % error

10,069

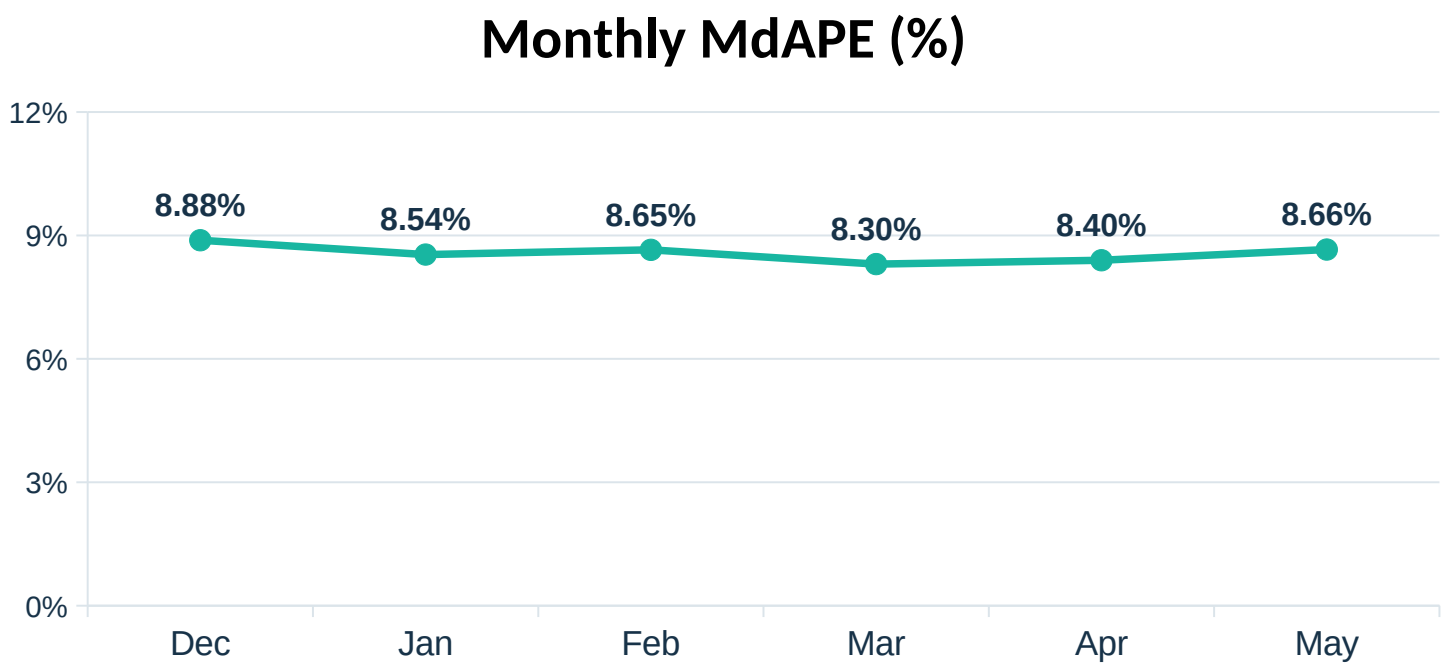
Held-out homes

December 2025–May 2026. Lower error is better. Reference baselines: earlier original-data run; identical homes and labels verified.

Refit on all original sales, export, and check

After validation: fit the final model → predict test homes → verify submission_off_market.csv

Monthly median absolute percentage error



3,974

off-market predictions

Final fit: all 150,000 original training sales

- ✓ Every test ID exactly once
- ✓ Positive, finite prices
- ✓ Official CSV checker passed

submission_off_market.csv

Validation is not the hidden test. Future market conditions remain a limitation.